

Numerical methods

Lecture 7: Nonlinear equations I.

Levente Lócsi

ELTE IK

- ① Bolzano's theorem, bisection method
- ② Fixed-point iterations
- ③ Order of convergence

Problem

Find the roots (zeros) of a nonlinear function $f \in \mathbb{R} \rightarrow \mathbb{R}$.
($\exists?$, 1, more?)

$$f(x^*) = 0, \quad x^* = ?$$

Can be equivalently reformulated (usually):

Find a fixed-point of a nonlinear function $\varphi \in \mathbb{R} \rightarrow \mathbb{R}$.

$$x^* = \varphi(x^*), \quad x^* = ?$$

① Bolzano's theorem, bisection method

② Fixed-point iterations

③ Order of convergence

Recall Analysis. . .

Theorem: Bolzano's theorem

If $f \in C[a; b]$ and $f(a) \cdot f(b) < 0$, then $\exists x^* \in (a; b) : f(x^*) = 0$.

Remarks:

- $a, b \in \mathbb{R}$, $a < b$, $[a; b]$ closed interval,
- $C[a; b]$: set of continuous functions on $[a; b]$,
- $f(a) \cdot f(b) < 0$: the sign of $f(a)$ and $f(b)$ are different
- exists a root in the $(a; b)$ (open) interval

Proof. (Bolzano's theorem): with the bisection method

- ① Let $x_0 := a$, $y_0 := b$.
- ② Repeat:
 - Let $z_k := \frac{1}{2}(x_k + y_k)$, midpoint of the interval.
 - If $f(x_k) \cdot f(z_k) < 0$, then $x_{k+1} := x_k$, $y_{k+1} := z_k$.
 - If $f(x_k) \cdot f(z_k) > 0$, then $x_{k+1} := z_k$, $y_{k+1} := y_k$.
- ③ Stop if
 - equality holds: $x^* = z_k$, or
 - the desired precision ε is reached: $x^* \in (x_k, y_k)$ with

$$y_k - x_k = \frac{b - a}{2^k} < \varepsilon.$$

... see Analysis ...



Remark:

- Equality is rarely met.
- For the detailed discussion of the convergence of sequences (x_k) and (y_k) see Analysis. . .
- **Error estimates:**

$$|x_k - x^*| < \frac{b-a}{2^k}, \quad |y_k - x^*| < \frac{b-a}{2^k},$$

$$|s_k - x^*| < \frac{b-a}{2^{k+1}}.$$

- This is a *slow* method.

Example

Find a root of the polynomial $P(x) = x^3 + 3x - 2$ with precision 0.1. How many steps are required?

Try the interval $[0; 1]$.

$$p(x) = x^3 + 3x - 2$$

$$p(0) = -2 < 0, \quad p(1) = 2 > 0, \quad p \in C[0, 1] \quad \Rightarrow \\ \Rightarrow (\text{Bolzano}) \quad \exists x^* \in (0, 1) : p(x^*) = 0.$$

$$\bullet \quad x_0 = 0, \quad y_0 = 1 \quad [0, 1] \quad z_0 = \frac{x_0 + y_0}{2} = \frac{0 + 1}{2} = \frac{1}{2}$$

$$p(z_0) = p(1/2) = \frac{1}{8} + 3 \cdot \frac{1}{2} - 2 = \frac{1 + 12 - 16}{8} = -\frac{3}{8} < 0$$

$$\bullet \quad x_1 = z_0 = \frac{1}{2}, \quad y_1 = y_0 = 1 \quad \left[\frac{1}{2}, 1\right] \quad z_1 = \frac{x_1 + y_1}{2} = \frac{\frac{1}{2} + 1}{2} = \frac{3}{4}$$

$$p(z_1) = p(3/4) = \frac{27}{64} + 3 \cdot \frac{3}{4} - 2 = \frac{27 + 144 - 128}{64} = \frac{43}{64} > 0$$

$$\bullet \quad x_2 = x_1 = \frac{1}{2}, \quad y_2 = z_1 = \frac{3}{4} \quad \left[\frac{1}{2}, \frac{3}{4} \right] \quad z_2 = \frac{x_2 + y_2}{2} = \frac{\frac{1}{2} + \frac{3}{4}}{2} = \frac{5}{8}$$

$$\rho(z_2) = \rho(5/8) = \dots$$

$$x^* \in [x_2, y_2] = \left[\frac{1}{2}, \frac{3}{4} \right], \quad x^* \approx z_2 = \frac{5}{8},$$

$$y_2 - x_2 = \frac{b-a}{2^2} = \frac{1}{4}, \quad |x^* - z_2| \leq \frac{b-a}{2^{2+1}} = \frac{1}{8}.$$

$$|x^* - z_k| \leq \frac{b-a}{2^{k+1}} \leq \frac{1}{10}, \quad (1-0) \cdot 10 \leq 2^{k+1}, \quad k+1 \geq 4, \quad k \geq 3.$$

Theorem: unique root

If $f \in C[a; b]$, $f(a) \cdot f(b) < 0$ and furthermore $f \in D(a; b)$ and $f' > 0$ (or < 0), then $\exists! x^* \in (a; b) : f(x^*) = 0$.

Proof. Bolzano's theorem $\Rightarrow \exists$ a root.

Furthermore f is strictly monotonic $\Rightarrow$ the root is unique.



① Bolzano's theorem, bisection method

② Fixed-point iterations

③ Order of convergence

Idea: for the roots of nonlinear functions:

$$\begin{aligned} f(x) = 0 & \iff x = \varphi(x) \\ x_{k+1} &= \varphi(x_k) = \dots \end{aligned}$$

Can be also applied for linear systems
($\rightsquigarrow$ iteration methods for SLE).

$$\begin{aligned} Ax = b & \iff x = Bx + c \\ x^{(k+1)} &= \varphi(x^{(k)}) = B \cdot x^{(k)} + c \end{aligned}$$

Definition: fixed-point

The point $x^* \in \mathbb{R}$ is called a *fixed-point* of the function (map) $\varphi: \mathbb{R} \rightarrow \mathbb{R}$, if $x^* = \varphi(x^*)$.

Definition: contraction

The map $\varphi: [a; b] \rightarrow \mathbb{R}$ is a *contraction* on $[a; b]$, if $\exists q \in [0, 1)$, such that

$$|\varphi(x) - \varphi(y)| \leq q \cdot |x - y|, \quad \forall x, y \in [a; b].$$

Remarks:

- contraction $\approx$ bringing things closer...
- q : contraction coefficient
- similarly in $\mathbb{R}^n$: $|\cdot| \rightsquigarrow \|\cdot\|$; $[a; b] \rightsquigarrow \mathbb{R}^n$

Proposition

If ① $\varphi: [a; b] \rightarrow \mathbb{R}$ function, $\varphi \in C^1[a; b]$ and

$$\textcircled{2} |\varphi'(x)| < 1 \quad (\forall x \in [a; b]),$$

then φ is a contraction on $[a; b]$.

Remarks:

- C^1 : one times continuously differentiable.
- Being a contraction can depend on the interval.

Proof: By the Lagrange mean value theorem.

$$q := \max_{x \in [a; b]} |\varphi'(x)| < 1$$

$$\forall x, y \in [a; b] \ (x < y) : \exists \xi \in (x; y) :$$

$$|\varphi(x) - \varphi(y)| = |\varphi'(\xi)| \cdot |x - y| \leq q \cdot |x - y|. \quad \square$$

Brouwer's fixed-point theorem

Theorem: Brouwer's fixed-point theorem

If $\varphi: [a; b] \rightarrow [a; b]$, $\varphi \in C[a; b]$, then $\exists x^* \in [a; b] : \varphi(x^*) = x^*$.

Proof. $\varphi(a), \varphi(b) \in [a, b]$, define $g(x) := x - \varphi(x)$

- $g(a) = a - \varphi(a) \leq 0$, $g(b) = b - \varphi(b) \geq 0$.
- If one of them is 0: a or b is a root of g , a fixed-point of φ .
- None of them is 0: using Bolzano's theorem ($g(a) \cdot g(b) < 0$):
 $\exists x^* \in [a, b] : g(x^*) = 0$, i.e. $x^* = \varphi(x^*)$.

Visually?

Theorem: Banach's fixed-point theorem for $[a; b]$

If the function $\varphi: [a; b] \rightarrow [a; b]$ is a contraction on $[a; b]$ with coefficient q , then

- 1 $\exists! x^* \in [a; b] : x^* = \varphi(x^*)$, i.e. a unique fixed-point exists,
- 2 $\forall x_0 \in [a; b]$ the sequence $x_{k+1} = \varphi(x_k)$, $k \in \mathbb{N}_0$ is convergent and $\lim_{k \rightarrow \infty} x_k = x^*$,
- 3 the below error estimates hold:
 - $|x_k - x^*| \leq q^k \cdot |x_0 - x^*| \leq q^k(b - a)$,
 - $|x_k - x^*| \leq \frac{q^k}{1 - q} \cdot |x_1 - x_0|$.

Proof. Skipped.



Can be similarly stated (and proved) in case of $\mathbb{R}^n$ too.

Corollary: sufficient condition for convergence

If ❶ $\varphi: [a; b] \rightarrow [a; b]$,

❷ $\varphi \in C^1[a; b]$ and

❸ $|\varphi'(x)| < 1 \quad (\forall x \in [a; b])$,

then the iteration $x_{k+1} = \varphi(x_k)$ is convergent $\forall x_0 \in [a; b]$.

Remark. The sequence may be convergent also if at some point(s) $|\varphi'| \geq 1$ holds. (It is not a necessary condition.)

Theorem: Local fixed-point theorem

Consider a function $\varphi: [a; b] \rightarrow \mathbb{R}$. If

- 1 $\varphi \in C^1[a; b]$, and
- 2 $\exists \xi \in [a; b]$ and $\delta > 0$, such that

$$|\varphi'(x)| \leq q < 1 \quad \forall x \in [\xi - \delta; \xi + \delta] \subset [a; b],$$

- 3 and $\exists r: 0 < r \leq \delta$, such that $|\varphi(\xi) - \xi| \leq (1 - q)r$
(i.e. ξ is a good approximation of the fixed-point),

then φ is a contraction on $[\xi - r; \xi + r]$ and

$$\forall x \in [\xi - r; \xi + r]: \varphi(x) \in [\xi - r; \xi + r].$$

Proof.

- From the conditions of the theorem it follows that φ is a contraction on the interval $[\xi - \delta; \xi + \delta]$.
- Then φ is also a contraction (with the same coefficient) on the $[\xi - r; \xi + r] \subset [\xi - \delta; \xi + \delta]$ subinterval too.
- For any arbitrary $x \in [\xi - r; \xi + r]$:

$$\begin{aligned} |\varphi(x) - \xi| &= |\varphi(x) - \varphi(\xi) + \varphi(\xi) - \xi| \leq \\ &\leq |\varphi(x) - \varphi(\xi)| + |\varphi(\xi) - \xi| \leq \\ &\leq q \cdot \underbrace{|x - \xi|}_{\leq r} + (1 - q) \cdot r = r \end{aligned}$$

Thus φ maps to the interval $[\xi - r; \xi + r]$.



Corollary

If the conditions of the local fixed-point theorem are satisfied, then

- ① $\exists! x^* \in [\xi - r; \xi + r] : x^* = \varphi(x^*)$, i.e. a unique fixed-point exists,
- ② $\forall x_0 \in [\xi - r; \xi + r]$ the sequence $x_{k+1} = \varphi(x_k)$, $k \in \mathbb{N}_0$ is convergent and $\lim_{k \rightarrow \infty} x_k = x^*$,
- ③ the below error estimates hold:
 - $|x_k - x^*| \leq q^k \cdot |x_0 - x^*| \leq q^k(b - a)$,
 - $|x_k - x^*| \leq \frac{q^k}{1 - q} \cdot |x_1 - x_0|$.

Proof. Actually the conditions of the Banach fixed-point theorem are fulfilled for the interval $[\xi - r; \xi + r]$. □

Example 1

Choose a number between 0 and 1, and keep squaring it repeatedly (e.g. keep pressing the x^2 button on a calculator). What happens after many iterations?

Actually we were approximating the solution of the equation $x = x^2$ with the iteration

$$x_{k+1} = x_k^2.$$

It has two fixed points: 0 and 1, but

- $\lim(x_k) = 0$ for $0 \leq x_0 < 1$,
- $\lim(x_k) = 1$ for $x_0 = 1$.

(Note that the condition of Banach's theorem does not hold.)

Example 2

Choose a number between 0 and 1, and keep taking its square root (e.g. keep pressing the $\sqrt{x}$ button on a calculator). What happens after many iterations?

Actually we were approximating the solution of the equation $x = \sqrt{x}$ with the iteration

$$x_{k+1} = \sqrt{x_k}.$$

It has two fixed points: 0 and 1, but

- $\lim(x_k) = 0$ for $x_0 = 0$,
- $\lim(x_k) = 1$ for $0 < x_0 \leq 1$.

Example 3

Choose a number between 0 and 1, and keep taking its cosine (e.g. keep pressing the $\cos(x)$ button on a calculator). What happens after many iterations?

Actually we were approximating the solution of the equation $x = \cos x$ on the interval $[0; 1]$ with the iteration

$$x_{k+1} := \cos(x_k), \quad x_0 \in [0, 1].$$

Is the solution unique? Is this sequence convergent? Give an error estimate! How many steps are required to find the solution with precision 0.0001?

$$x_{k+1} = \cos x_k, \quad x_0 := 0; \quad x_1 = \cos 0 = 1, \quad x_2 = \cos 1 \approx 0.54, \quad \dots$$

$$\bullet \quad \varphi : [0, 1] \rightarrow [0, 1] \quad \varphi(x) = \cos x,$$

$$\varphi(0) = 1, \quad \varphi(1) \in (0, 1), \quad \varphi'(x) = -\sin x \leq 0 \quad (x \in [0, 1]) \quad \Rightarrow \quad \checkmark$$

$$\bullet \quad |\varphi'| < 1 \quad |\varphi'(x)| = |-\sin x| = \sin x \leq \sin 1 < 1 \quad (x \in [0, 1])$$

$$q := \sin 1 \approx 0.84 \quad \checkmark$$

$$\bullet \Rightarrow (\text{Banach}) \quad x_{k+1} = \cos x_k \text{ conv. } \forall x_0 \in [0, 1], \exists! x^*.$$

$$|x_k - x^*| \leq q^k \cdot |x_0 - x^*| = q^k \cdot \underbrace{|b - a|}_1 = q^k \leq 10^{-4}$$

$$k \cdot \ln q \leq -4 \cdot \ln 10, \quad k \geq \frac{-4 \ln 10}{\ln \sin 1} \approx 53.366, \quad k \geq 54$$

Example 4

Examine the behavior of the iteration

$x_0 \in [0, 1]$, $x_{k+1} = \alpha \cdot x_k(1 - x_k)$ (the logistic map) for different $\alpha \in [0, 4]$ parameters.

Remark. Usually not a contraction. We may easily arrive at the basic examinations about bifurcations of difference equations and the theory of chaos. . .

- ① Bolzano's theorem, bisection method
- ② Fixed-point iterations
- ③ Order of convergence

Definition: order of convergence

The convergent sequence (x_k) with limit x^* is said to have *order of convergence* p , if $\exists c \in (0; +\infty) \subset \mathbb{R}$, such that

$$\lim_{k \rightarrow \infty} \frac{|x_{k+1} - x^*|}{|x_k - x^*|^p} = c.$$

Remarks.

- p is unique, $p \geq 1$,
- p is not necessarily integer (secant method: $p = \frac{1+\sqrt{5}}{2}$.)
- $p = 1$: linear or first order convergence (with $c \leq 1$)
 $p = 2$: quadratic or second order convergence
- $p > 1$: superlinear convergence ($p < 1$: sublinear conv. is rare)

Remark continued.

- Practically at least p th order convergence can be stated as:

$$\exists K \in \mathbb{R}^+ : \forall k \in \mathbb{N}_0 : |x_{k+1} - x^*| \leq K \cdot |x_k - x^*|^p$$

- Fixed-point theorems did not state order of convergence. (Only that it is at least linear.)
- If $c = 0$, then the order of convergence is higher than q .
- If $c = \infty$, then the order of convergence is lower than q .

Example

What is the order of convergence of the following null-sequences?

$$\left(\frac{1}{n^2}\right); \quad \left(\frac{1}{2^n}\right); \quad (q^n) \ (|q| < 1); \quad \left(\frac{1}{2^{2^n}}\right);$$

$$\frac{\left| \frac{1}{(n+1)^2} - 0 \right|}{\left| \frac{1}{n^2} - 0 \right|^1} = \frac{n^2}{(n+1)^2} = \frac{n^2}{n^2 + 2n + 1} \longrightarrow 1 \in (0, +\infty) \Rightarrow p = 1$$

$$\frac{\left| \frac{1}{2^{n+1}} - 0 \right|}{\left| \frac{1}{2^n} - 0 \right|^2} = \frac{2^{2n}}{2^{n+1}} = 2^{n-1} \longrightarrow +\infty \notin (0, +\infty) \Rightarrow p < 2$$

$$\frac{\left| \frac{1}{2^{n+1}} - 0 \right|}{\left| \frac{1}{2^n} - 0 \right|^1} = \frac{2^n}{2^{n+1}} = \frac{1}{2} \longrightarrow \frac{1}{2} \in (0, +\infty) \Rightarrow p = 1$$

$$\frac{\left| \frac{1}{2^{2n+1}} - 0 \right|}{\left| \frac{1}{2^{2^n}} - 0 \right|^2} = \frac{2^{2 \cdot 2^n}}{2^{2^{n+1}}} = 1 \longrightarrow 1 \in (0, +\infty) \Rightarrow p = 2$$

What does the order of convergence mean in numbers? ($\sqrt{2}$)

① $p = 1, |x_{k+1} - x^*| \leq K \cdot |x_k - x^*|^1$

1.414184570312500

1.414245605468750

1.414215087890625

In each step approximately one more digit is accurate.

② $p = 2, |x_{k+1} - x^*| \leq K \cdot |x_k - x^*|^2$

1.416666666666667

1.414215686274510

1.414213562374690

In each step approx. twice as many digits are accurate.

Theorem: iterations with order of convergence p

- If
- ① $\varphi: \mathbb{R} \rightarrow \mathbb{R}$, $\varphi \in C^p[a; b]$ and
 - ② the sequence $x_{k+1} = \varphi(x_k)$ is convergent having limit x^* ,
 - ③ $\varphi'(x^*) = \dots = \varphi^{(p-1)}(x^*) = 0$, but $\varphi^{(p)}(x^*) \neq 0$,
- then the order of convergence is p and the error estimate

$$|x_{k+1} - x^*| \leq \frac{M_p}{p!} |x_k - x^*|^p,$$

holds with $M_p = \max_{\xi \in [a; b]} |\varphi^{(p)}(\xi)|$.

Proof. Write the Taylor-polynomial of the function φ around x^* with the remainder term.

$\exists \xi \in (x, x^*)$ (or (x^*, x)) :

$$\begin{aligned}\varphi(x) = & \varphi(x^*) + \varphi'(x^*)(x - x^*) + \cdots + \frac{\varphi^{(p-1)}(x^*)}{(p-1)!}(x - x^*)^{p-1} + \\ & + \frac{\varphi^{(p)}(\xi)}{p!}(x - x^*)^p\end{aligned}$$

Substitute $x = x_k$, using the zero derivatives too. ($\exists \xi_k$):

$$x_{k+1} = \varphi(x_k) = \underbrace{\varphi(x^*)}_{x^*} + \frac{\varphi^{(p)}(\xi_k)}{p!}(x_k - x^*)^p$$

Proof continued. Reorder to

$$x_{k+1} - x^* = \frac{\varphi^{(p)}(\xi_k)}{p!} (x_k - x^*)^p.$$

$$\lim_{k \rightarrow \infty} \frac{|x_{k+1} - x^*|}{|x_k - x^*|^p} = \lim_{k \rightarrow \infty} \frac{|\varphi^{(p)}(\xi_k)|}{p!} = \frac{|\varphi^{(p)}(x^*)|}{p!} \neq 0.$$

Thus the order of convergence for (x_k) is p .

Examine the errors of x_{k+1} and x_k .

$$|x_{k+1} - x^*| = \frac{|\varphi^{(p)}(\xi_k)|}{p!} \cdot |x_k - x^*|^p \leq \frac{M_p}{p!} |x_k - x^*|^p,$$

with $M_p = \max_{\xi \in [a,b]} |\varphi^{(p)}(\xi)|$.



Corollary

- If
- ① $\varphi: [a; b] \rightarrow [a; b]$ is a contraction,
 - ② x^* is a fixed-point of φ , and
 - ③ $\varphi'(x^*) = \dots = \varphi^{(p-1)}(x^*) = 0$, but $\varphi^{(p)}(x^*) \neq 0$,

then

- ① the fixed-point is unique,
- ② $\forall x_0 \in [a; b]$ the sequence $x_{k+1} = \varphi(x_k)$, $k \in \mathbb{N}_0$ is convergent and $\lim_{k \rightarrow \infty} x_k = x^*$,
- ③ furthermore the below error estimate holds:
$$|x_{k+1} - x^*| \leq \frac{M_p}{p!} |x_k - x^*|^p.$$

Proof. Follows from the Banach fixed-point theorem and the theorem about p th order iterations.



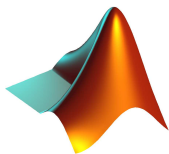
Example

Formulate fixed-point iterations for the solution of the equation $x^3 - x - 1 = 0$ and prove convergence.

a $x = x^3 - 1,$

b $x = \sqrt[3]{x+1}.$

One is convergent, the other one is divergent. Which is which?
Give an appropriate interval (with reasons).



- 1 Bisection method: calculation, visualization.
- 2 Fixed-point iterations via the example of $x = \cos(x)$.
- 3 The behavior of the logistic map (just for fun).